



B.Tech. Degree VIII Semester Special Supplementary Examination July 2025

ME 19-205-0810 OPERATIONS RESEARCH (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Solve problems using optimization methods.
 CO2: Develop mathematical skills to approach real life industrial problems.
 CO3: Inculcate ideas to solve problems on Transportation and Queueing models.
 CO4: Import and analyse case studies to solve future problems.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create

PO – Programme Outcome

PART A a, b, c, d, e, f
 PART B 2, 4, 6

PART A (Answer ALL questions)

(8 × 3 = 24)				
	Marks	BL	CO	PO
I. (a) What are the major limitations of using Linear Programming models in practical decision-making? Support your answer with examples.	3	L2	1	1,3
(b) Briefly explain the phases of an Operations Research study.	3	L2	1	1,3
(c) What are the optimality conditions in the simplex method?	3	L2	2	1,6
(d) Distinguish between Phase I and Phase II in the Two-phase method.	3	L2	2	1,6
(e) Compare assignment problems and transportation problems based on any three parameters.	3	L2	3	1,3
(f) Define an unbalanced assignment problem. How do you handle unbalanced problems in assignment problems?	3	L2	3	1,3
(g) State Bellman's principle of optimality with a suitable example.	3	L2	3	1,7
(h) Write the assumptions of the generalized Poisson queueing model.	3	L2	3	1,7

PART B

(4 × 12 = 48)				
II. Minimize $Z = 3x_1 + 2x_2$ solve by graphically. Subject to $5x_1 + x_2 \geq 10$ (0,10) (2,0) $x_1 + x_2 \geq 6$ (0,6) (6,0) $x_1 + 4x_2 \geq 12$ (0,3) (12,0) $x_1, x_2 \geq 0$	12	L3	1	1,3

OR

III. (a) Discuss the role of Operations Research in modern industry, giving at least three examples of application areas.	6	L2	1	1,3
(b) Explain any two Operations Research models in detail.	6	L2	1	1,3

(P.T.O.)

Marks	III	CO	
12	1.3	1	1.3

IV. Use simplex method to solve the LPP.

$$\text{Max } Z = 3x_1 + 2x_2$$

$$\text{Subject to } x_1 + x_2 \leq 4$$

$$x_1 - x_2 \leq 2$$

$$x_1, x_2 \geq 0$$

OR

V. (a) Explain the procedure of the Big M method and explain how it handles artificial variables.

(b) With the help of a suitable example, explain the economic interpretation of duality in Linear Programming?

6	1.2	2	1.6
6	1.2	2	1.6
12	1.3	1	1.3

VI.

Solve the following assignment problem for minimizing cost.

From	To				
	I	II	III	IV	V
A	1	3	2	3	6
B	2	4	3	1	5
C	5	6	3	4	6
D	3	1	4	2	2
E	1	5	6	5	4

OR

VII. Solve the given transportation problem by North West Corner Rule and Least Cost Method.

12	L3	1	1.3
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	D1	D2	D3	D4	supply
S1	1	2	3	4	16
S2	4	3	2	0	8
S3	0	2	2	1	20
Demand	14	10	8	12	

(Continued)

		Marks	BL	CO	PO
VIII. (a)	Customers arrive at a one-window drive-in bank according to Poisson distribution with mean 10 per hour. Service time per customer is exponential with mean five minutes. The space in front of the window including that for the serviced car can accommodate a maximum of three cars. Others can wait outside this space.	6	L3	1	1,7
	(i) What is the probability that an arriving customer can drive directly to the space in front of the window?				
	(ii) What is the probability that an arriving customer will have to wait outside the indicated space?				
	(iii) How long is an arriving customer expected to wait before starting service?				
(b)	People arrive at a theatre ticket centre in a Poisson distributed arrival rate of 25 per hour. Serve time is constant at two minutes. Calculate	6	L3	1	1,7
	(i) The mean number in the waiting line.				
	(ii) The mean waiting time.				
	(iii) Utilization factor.				
OR					
IX. (a)	Describe the differences between forward and backward recursive approaches in dynamic programming, with an example.	6	L2	2	1,7
(b)	A company can invest in two projects, A and B. Total available investment = ₹10 lakhs.	6	L3	1	1,7
	(i) Project A: Requires ₹5 lakhs, expected return = ₹8 lakhs.				
	(ii) Project B: Requires ₹10 lakhs, expected return = ₹14 lakhs.				
	Formulate and solve this problem using dynamic programming to maximize return.				

Blooms's Taxonomy Levels
L2 – 45%, L3 – 55%.
